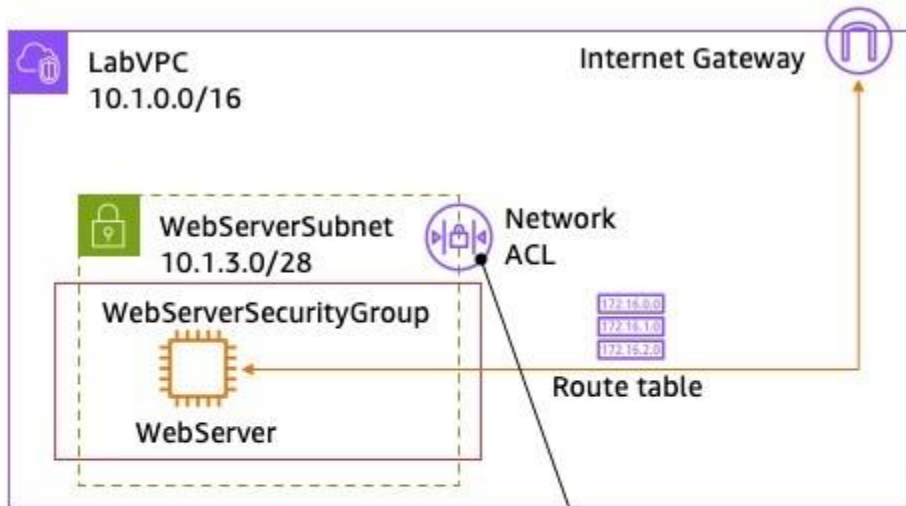


Lab: Secure the Web Server Subnet with a network ACL

In this lab, you will configure a network access control list (ACL) to secure the subnet where the web server is running. The network ACL will provide an additional layer of security beyond the security group that you already configured.

An example of network ACL architecture is shown in the following diagram:



NACL		
Rule number	Type	Allow/Deny
100	SSH (22)	Allow
90	HTTP (80)	Allow

Add and delete rules

When you add or delete a rule from an ACL, any subnets that are associated with the ACL are subject to the change. You don't have to terminate and relaunch the instances in the subnet. The changes take effect after a short period.

Check the associated Network ACL of the Subnet of your VPC.

1. Open the Amazon **VPC console**.
2. In the navigation pane, choose **Network ACLs**.
3. In the details pane, choose either the **Inbound Rules** or **Outbound Rules** tab, depending on the type of rule that you need to add, and then **choose Edit**.
4. In Rule #, enter a rule number (for example, 90). The rule number must not already be in use in the network ACL. NACL process the rules in order, starting with the lowest number.
5. Select a rule from the Type list. For example, to add a rule for HTTP, choose HTTP.
6. In the Source or Destination field (depending on whether this is an inbound or outbound rule), enter the CIDR range that the rule applies to for example 0.0.0.0/0 for all source.
7. From the Allow/Deny list, select ALLOW to allow the specified traffic or DENY to deny the specified traffic.
8. (Optional) To add another rule, choose Add another rule, and repeat steps 4 to 9 as required.
9. When you are done, choose Save.
10. To test whether you can access port 80 on the WebServer instance, modify the 90 rule from Allow to Deny. Then, access web server
11. The connection fails or times out, which indicates that access is no longer allowed.

Analysis: Network ACL rules can block network access even when security group inbound rules allow access. Security groups can block network traffic at the instance level; however, Network ACLs can block traffic at the subnet level and thus offer an additional layer of security.

12. Modify rule number 90 to allow connections on port 80.

Now, you should see the "Hello world from Web Server!" message.